M2: Initial Implementation
Customer Segmentation and Market Basket Analysis

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***Abstract*—This milestone summarizes the initial implementation of a data mining pipeline using the UCI Online Retail dataset. The goal of this stage is to explore the dataset, perform preprocessing, and apply initial mining techniques to discover patterns in customer transactions. Exploratory Data Analysis (EDA) reveals key characteristics of the dataset, while preprocessing improves data quality by removing invalid transactions. Association rule mining is then applied to identify frequently co-purchased products. The findings provide early insights into customer purchasing behavior and establish the foundation for more advanced analysis in the next milestone.**

I. Key Exploratory Data Analysis Insights

The Online Retail dataset contains approximately 541,909 transaction records with eight attributes describing purchase behavior, including invoice number, product code, quantity, unit price, and customer information.

Initial EDA revealed several important patterns. First, the distribution of purchase quantities is highly skewed. Most transactions involve small quantities, while a small number of transactions contain bulk purchases. This long-tail distribution is common in retail datasets.

Second, the majority of customers are located in the United Kingdom. Other countries appear in the dataset but represent a much smaller portion of transactions. This indicates that the retailer primarily serves a domestic customer base.

Third, product prices are generally low, with most items costing only a few units of currency. However, several outliers exist with unusually high prices or quantities. These observations highlight the need for careful data cleaning before applying mining algorithms.

II. Data Preprocessing Decisions

Several preprocessing steps were applied to improve the quality of the dataset.

Rows containing missing values were removed because incomplete records can introduce noise into the analysis. In particular, some transactions do not contain customer identifiers, which limits their usefulness for later customer-based analysis.

Transactions representing product returns were also removed. These rows contain negative quantities and do not represent actual purchases. Including them would distort patterns of customer buying behavior.

Additionally, rows with zero or negative prices were filtered out to avoid unrealistic transaction values. After cleaning the data, a new feature called *TotalPrice* was created by multiplying quantity and unit price. This feature represents the monetary value of each transaction item and is useful for later analysis.

These preprocessing steps significantly improved the reliability of the dataset.

III. Initial Data Mining Results

The first mining technique applied in this milestone is association rule mining using the Apriori algorithm. The dataset was converted into a transactional format where each invoice corresponds to a set of purchased items.

Frequent itemsets were identified based on minimum support thresholds. Association rules were then generated using three metrics: support, confidence, and lift.

Several meaningful product associations were discovered. For example, certain decorative or gift-related items frequently appear together within the same transaction. High lift values for these rules suggest that the relationships between these products are not random but reflect common purchasing patterns.

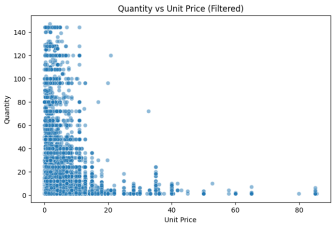

Fig. 1. The scatter plot shows the relationship between unit price and purchase quantity. Most transactions involve low priced items and small quantities. There is no strong linear relationship between price and quantity, suggesting that customers do not necessarily buy more when prices are lower.

	antecedents	consequents	support	confidence	lift
31	PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCE	0.021045	0.701439	24.027846
27	GREEN REGENCY TEACUP AND SAUCE	PINK REGENCY TEACUP AND SAUCER	0.021045	0.720887	24.027846
28	PINK REGENCY TEACUP AND SAUCER,	GREEN REGENCY TEACUP AND SAUCE	0.021045	0.894495	23.989564
30	GREEN REGENCY TEACUP AND SAUCE	PINK REGENCY TEACUP AND SAUCER,	0.021045	0.564399	23.989564
7	PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCE	0.024822	0.827338	22.188466
8	GREEN REGENCY TEACUP AND SAUCE	PINK REGENCY TEACUP AND SAUCER	0.024822	0.665702	22.188466
29	GREEN REGENCY TEACUP AND SAUCE	ROSES REGENCY TEACUP AND SAUCE	0.021045	0.847826	20.0663
22	PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCE	0.023527	0.784173	18.559754
21	ROSES REGENCY TEACUP AND SAUCE	PINK REGENCY TEACUP AND SAUCER	0.023527	0.556833	18.559754
11	GREEN REGENCY TEACUP AND SAUCE	ROSES REGENCY TEACUP AND SAUCE	0.029193	0.782923	18.530185

Fig. 2. highlights the most valuable association rules discovered by the Apriori algorithm. This table allows managers to easily identify which product combinations to promote, bundle, or cross-sell.

These results demonstrate that the dataset contains valuable information that can be used to understand product relationships and customer behavior.

IV. Plan for Milestone 3

The next milestone will expand the analysis by applying additional data mining techniques.

Customer segmentation will be performed using clustering algorithms such as K-Means and DBSCAN in order to identify groups of customers with similar purchasing patterns. These clusters may represent different types of shoppers, such as frequent buyers or high-value customers.

In addition, anomaly detection techniques such as Isolation Forest and Local Outlier Factor (LOF) will be used to identify unusual transactions or customers. These methods may reveal potential fraud, data errors, or rare purchasing behaviors.

Compared with the initial proposal in Milestone 1, the overall analysis plan remains largely the same. However, the results from EDA suggest that additional attention should be given to outliers and extreme transaction values during future analysis.

V. Project Structure

The project repository is organized as follows:

```
CS4412-Retail-Analysis-Project-JL/

 README.md

 data/
    raw/
       Online Retail.xlsx

 notebooks/
    M1_Proposal.ipynb
    M2_Initial_Implementation.ipynb
 src
 figures/
    CorrelationHeatmap.png
    FilteredQuantityDistribution.png
    MonthlySalesTrend.png
   QuantityvsUnitPrice(Filtered).png
    Top10CountriesbyQuantitySold.png
    Top10ProductsbyQuantitySold.png
    UnitPriceDistribution.png
 docs/
    M1-Proposal.pdf
    M2-Initial Implementation.pdf
 requirements.txt
```